

David Bedolla, PhD

Robotics Software Engineer | Manipulation, Robot Learning & Real-Time Systems

Portfolio | [LinkedIn](#) | [GitHub](#) | davidbedollamartinez@outlook.es | Montréal, QC, Canada

PROFESSIONAL PROFILE

PhD-level robotics software engineer with completed work in physical Kinova Gen3 control, mobile manipulation, ROS 2 and C++, teleoperation, force-aware interaction, multimodal sensing, and real-time exoskeleton deployment. Robot-learning experience includes Gaussian Process control models, neural-network inverse kinematics, and PyTorch reinforcement-learning experiments in simulation. I also adapted contact sensing across Gazebo and MuJoCo.

Learning from Demonstration for manipulation is my current development direction, supported by completed work in teleoperation, multimodal sensing, cross-simulator contact modeling, and model-based execution.

PROFESSIONAL EXPERIENCE

Associate Researcher

Mar 2026-Present | Lab INIT Robots, Montréal

- Deployed 400 Hz ROS 2 and C++ whole-body control on a physical 10-DoF AgileX and Kinova Gen3 mobile manipulator.
- Implemented analytical position targets and Pinocchio Jacobian-based joint velocities for Cartesian tracking, elbow reconfiguration, joint limits, and arm-base coordination.
- Integrated filtered odometry, world-frame tracking, command validation, safe state transitions, and PS4 teleoperation for repeatable physical experiments.
- Exercised the same architecture in Gazebo and MuJoCo; the analytical redundancy calculation runs in approximately 3 microseconds on an Intel i7 laptop.

Postdoctoral Researcher

Mar 2025-Mar 2026 | ÉTS, Montréal

- Reduced lower-limb robot control latency by 30 times, enabling torque control with lower tracking error on the target hardware.
- Improved ROS and Python scheduling for autonomous-vehicle control and reduced steering oscillations during testing.
- Developed symbolic dynamics and inverse-kinematics tools to accelerate modeling, controller integration, and validation.

Research Professor

Mar 2024-Mar 2025 | Universidad Tecnológica de la Mixteca, Mexico

- Developed and tested a 7-DoF robot optimization strategy that improved tracking performance by 20 percent.
- Defined mechanical and control requirements for a three-finger adaptive robotic gripper for daily-living tasks.

Research Assistant

Jan 2020-Dec 2023 | ÉTS, Montréal

- Deployed a 1-4 kHz LabVIEW RT/FPGA and C++ stack on a physical 7-DoF exoskeleton, including DAQ, actuator interfaces, safety logic, and trajectory execution.
- Integrated ATI Nano17 force/torque sensing and admittance-based task-space references; developed an external-force observer from motor current, dynamics, friction, and the Jacobian.
- Combined IMU, EMG, and depth-camera motion acquisition with sensor-guided references and a physical mirror-rehabilitation workflow.
- Deployed Gaussian Process residual modeling in robust MPC with sliding-mode compensation at approximately 387 microseconds per iteration on the physical exoskeleton.
- Achieved approximately 5 microseconds for closed-form inverse kinematics, 43 microseconds for learned inverse kinematics, and 210 microseconds for feedforward torque; reduced symbolic expressions by approximately 90 percent.

SELECTED ENGINEERING PROJECTS

Whole-Body Mobile Manipulation

Problem: hold a world-frame end-effector target while the base moves. **Contribution:** odometry transforms, arm-base coordination, redundancy handling, and teleoperation. **Hardware:** physical AgileX/Kinova. **Simulation:** Gazebo and MuJoCo. **Stack:** ROS 2, C++, Pinocchio. **Result:** 400 Hz. **Validation:** physical and simulation. [Video](#).

Cross-Simulator Tactile Sensor Adaptation

Problem: decouple tactile contact data from a single physics engine for whole-robot manipulation simulation. **Contribution:** adapted the open-source Lab-CORO TSF-85 approach for Gazebo and MuJoCo, connected tactile maps through ROS 2, and integrated whole-body control. **Simulation:** Gazebo and MuJoCo; upstream TSF-85 reference for Isaac Sim. **Stack:** ROS 2, Gazebo, MuJoCo, whole-body control. **Result:** tactile contact maps through the same ROS 2 interface in both target simulators. **Validation:** simulation. [Video](#) | [TSF-85 upstream](#) | [Apache-2.0](#).

Learning-Based Predictive Control and Force-Aware Rehabilitation

Problem: constrained exoskeleton tracking with model error and human interaction. **Contribution:** Gaussian Process residual modeling, robust MPC, force/torque sensing, admittance, and force observation. **Hardware:** physical 7-DoF ETS-MARSE. **Stack:** LabVIEW RT/FPGA, C++, GP/MPC. **Result:** 387 microseconds within a 1-4 kHz stack. **Validation:** physical. [Video](#).

URDF-Driven Trajectory Planning

Problem: reuse one planning interface across robot kinematics. **Contribution:** automatic joint controls, joint/Cartesian planning, inverse kinematics, limits, and trajectory analysis. **Simulation:** Kinova Gen3 model. **Stack:** C++, Qt, OpenGL, URDF, Ruckig. **Result:** inverse-kinematics tests up to 1 kHz. **Validation:** software/simulation. [Video](#).

TECHNICAL SKILLS

Programming: C, C++, Python, MATLAB, LabVIEW, VHDL

Robotics: ROS 2, Pinocchio, URDF, Cartesian control, redundancy resolution, whole-body control, mobile manipulation, teleoperation, motion planning

Robot Learning: PyTorch, reinforcement learning, Gaussian Processes, learning-based model predictive control, neural-network inverse kinematics

Interaction and Sensing: Force/torque sensing, admittance control, external-force observation, tactile-sensor simulation, IMU, EMG, depth cameras

Modeling and Control: Robot kinematics and dynamics, Recursive Newton-Euler Algorithm, numerical optimization, model predictive control, sliding-mode control, trajectory generation

Simulation: Gazebo, MuJoCo, Isaac Sim, Simulink, Simscape Multibody, LabVIEW digital twins

Real-Time Systems and Hardware: Real-time C++ control, LabVIEW RT/FPGA, DAQ systems, digital signal controllers, microcontrollers

Earlier Machine-Learning Experience: TensorFlow, convolutional neural networks, computer vision

EDUCATION

2020-2023 | PhD in Robotics - Learning-Based Upper-Limb Robotic Rehabilitation, ÉTS, Montréal

2014-2016 | M.Sc. in Robotics - Hardware-in-the-Loop Simulation of a Robotic Manipulator, Universidad Tecnológica de la Mixteca

2009-2014 | B.Eng. in Mechatronics - Universidad Tecnológica de la Mixteca; CENEVAL EGEL Outstanding distinction

SELECTED PUBLICATIONS

Bedolla-Martínez, D., Kali, Y., Saad, M., Ochoa-Luna, C., and Rahman, M. H. (2023). *Learning human inverse kinematics solutions for redundant robotic upper-limb rehabilitation*. Engineering Applications of Artificial Intelligence, 126, 106966. DOI: [10.1016/j.engappai.2023.106966](https://doi.org/10.1016/j.engappai.2023.106966).

Bedolla-Martínez, D., Kali, Y., Saad, M., Ochoa-Luna, C., and Rahman, M. H. (2023). *Robust MPC with Integral Super-Twisting for Trajectory Tracking of an Exoskeleton Robot Arm*. IEEE PEDS. DOI: [10.1109/PEDS57185.2023.10246735](https://doi.org/10.1109/PEDS57185.2023.10246735).